3	**B** $s_y = u_y t + \frac{1}{2} a_y t^2$ $500 = 0 + \frac{1}{2}(9.81)t^2$ → t = 10.1 s (time taken for bomb to drop to sea level) Horizontal distance travelled by bomb = 138 × 10.1 = 1394 m Horizontal distance travelled by ship = 12 × 10.1 = 121 m Release bomb when distance between plane and ship = 1394 – 121 = 1273 m OR relative speed between plane and ship = 138 – 12 = 126 m s^{-1} Distance = 126 × 10.1 = 1273 m
4	**D**